Experiment- 1: Getting Started - Tableau Workspace, Tableau terminologies, basic functionalities.

The **Tableau workspace** is the interface where you build visualizations (charts, dashboards, stories). When you open Tableau, you will see:

➢ **Tableau Desktop Workspace Components:**

- **Menu Bar**: File, Data, Worksheet options.
- **Toolbar**: Shortcuts for Save, Undo, Redo, Sort, etc.
- **Data Pane**: Shows your connected data (tables, fields, dimensions, measures).
- **Analytics Pane**: Drag analytics features like Trend Line, Forecast, etc.
- **Shelves**:
  - **Columns and Rows Shelves:** For placing fields to create the visualization.
  - **Filters Shelf:** Add filters to control data shown.
  - **Marks Card:** Control color, size, label, tooltip, shape.
- **View Area (Canvas):** Displays the actual chart or table.
- **Sheet Tabs:** Similar to Excel tabs – for switching between multiple worksheets, dashboards, or stories.

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2. Common Tableau Terminologies

Term	Description
Workbook	File that contains sheets (worksheets, dashboards, stories).
Worksheet	Single sheet where a chart or graph is built.
Dashboard	Collection of multiple views (charts) shown together.
Story	Sequence of sheets to tell a data narrative.
Data Source	The data connection (Excel, SQL, CSV, etc.).
Dimension	Categorical fields like names, dates, regions.
Measure	Numerical fields used for calculations (sales, profit).
Filter	Limits the data displayed in a view.
Mark	A point in the view, like a bar, circle, line point.
Calculated Field	Custom field created using formulas.
LOD Expression	Level of Detail – used to compute values at different data granularities.

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3. Basic Functionalities of Tableau

Feature	Description
Drag & Drop	Build visualizations by dragging fields into rows/columns or marks.
Chart Types	Create bar charts, line graphs, pie charts, maps, etc. automatically.

Feature	Description
🔍 **Filters**	Add filters to worksheets to refine the data shown.
🎨 **Marks Card**	Control color, size, label, tooltip of each chart element.
➕ **Create Calculated Fields**	Perform custom calculations (e.g., Profit Ratio = Profit / Sales).
□ **Aggregation**	Measures like SUM, AVG, MIN, MAX are auto-applied.
📌 **Show Me Panel**	Suggests suitable chart types based on selected fields.
📅 **Date Hierarchy**	Automatically drill into year → quarter → month → day.
□ **Analytics Pane**	Drag and drop Trend Lines, Reference Lines, Forecasts.
📤 **Publish to Tableau Server / Public**	Share visualizations with others online.

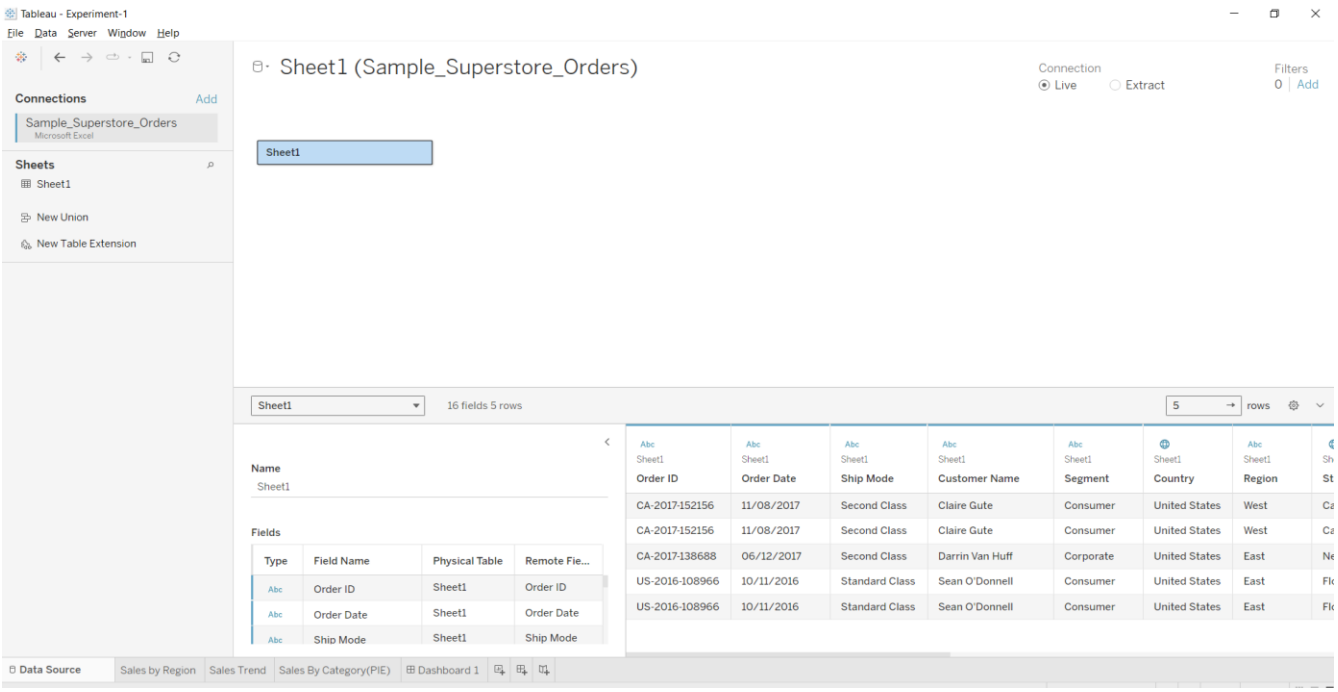
Example Workflow:

1. **Connect to Data** → Excel → Select sheet.
2. **Drag "Region" to Columns** and **"Sales" to Rows** → Bar chart appears.
3. **Drag "Category" to Color** → Bars split by product category.
4. **Add Filter for "Year"** → Only see selected year data.
5. **Create Dashboard** → Drag charts to dashboard layout.
6. **Publish or Export** to share your work.

Steps to Create the Tableau Workbook (.twbx)

Step 1. Open Tableau Desktop or Tableau Public

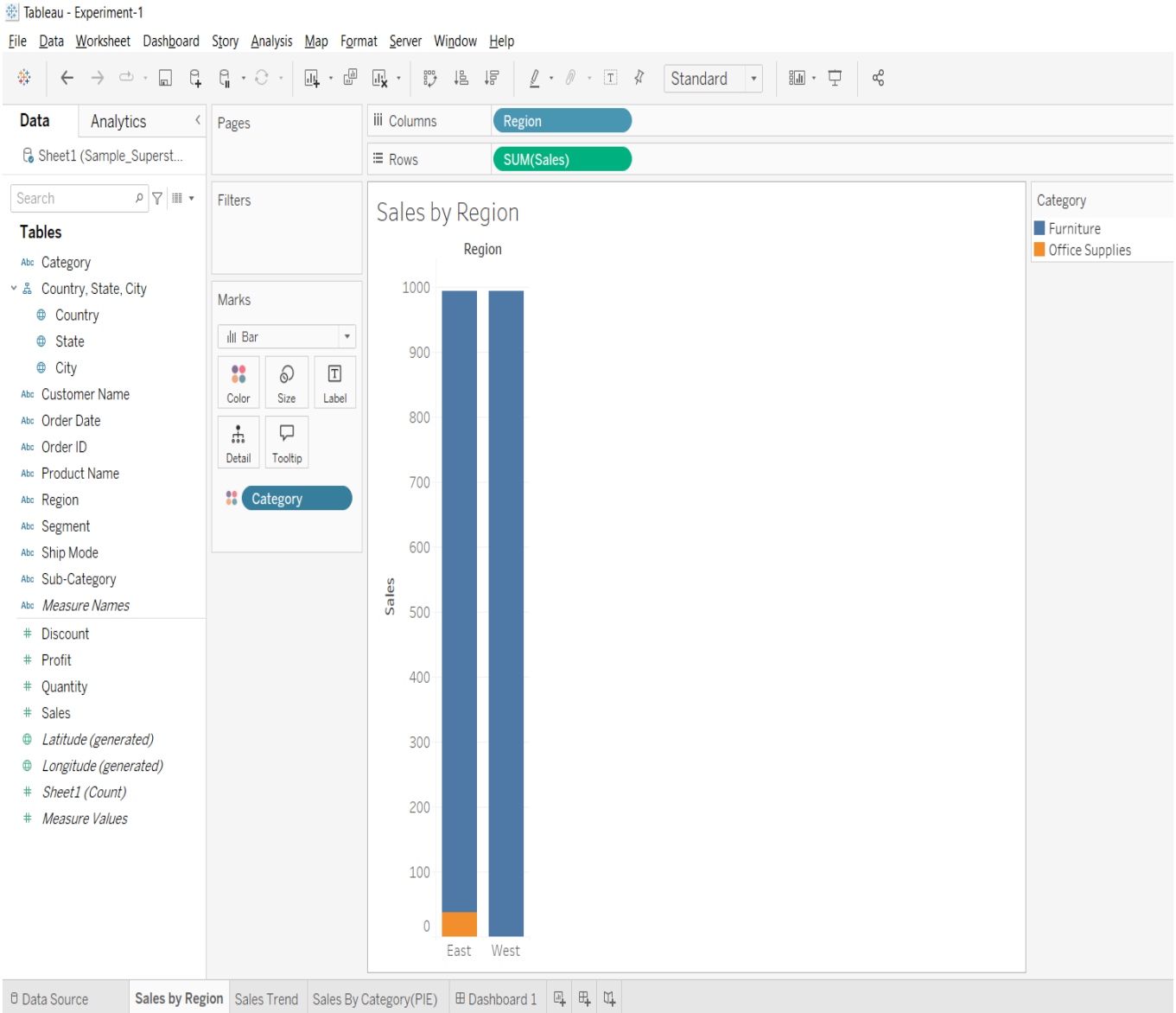
- Launch Tableau
- From the left panel, under **Saved Data Sources**, click on **Sample - Superstore**



Step 2. Create Three Basic Sheets

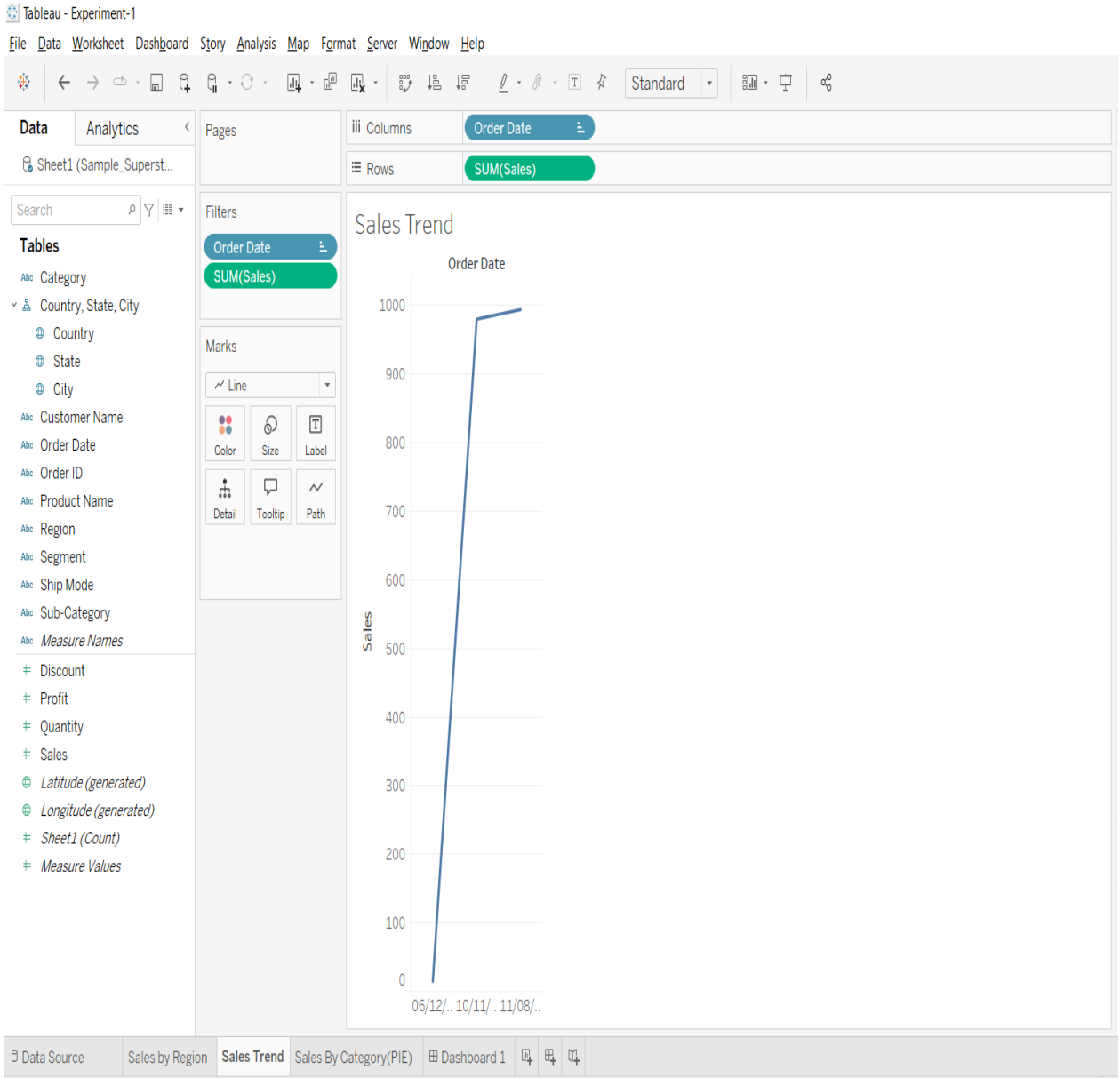
Sheet 1: Sales by Region

- Drag `Region` → Columns
- Drag `Sales` → Rows
- Drag `Category` → Color (on Marks card)
- Rename sheet to: `Sales by Region`



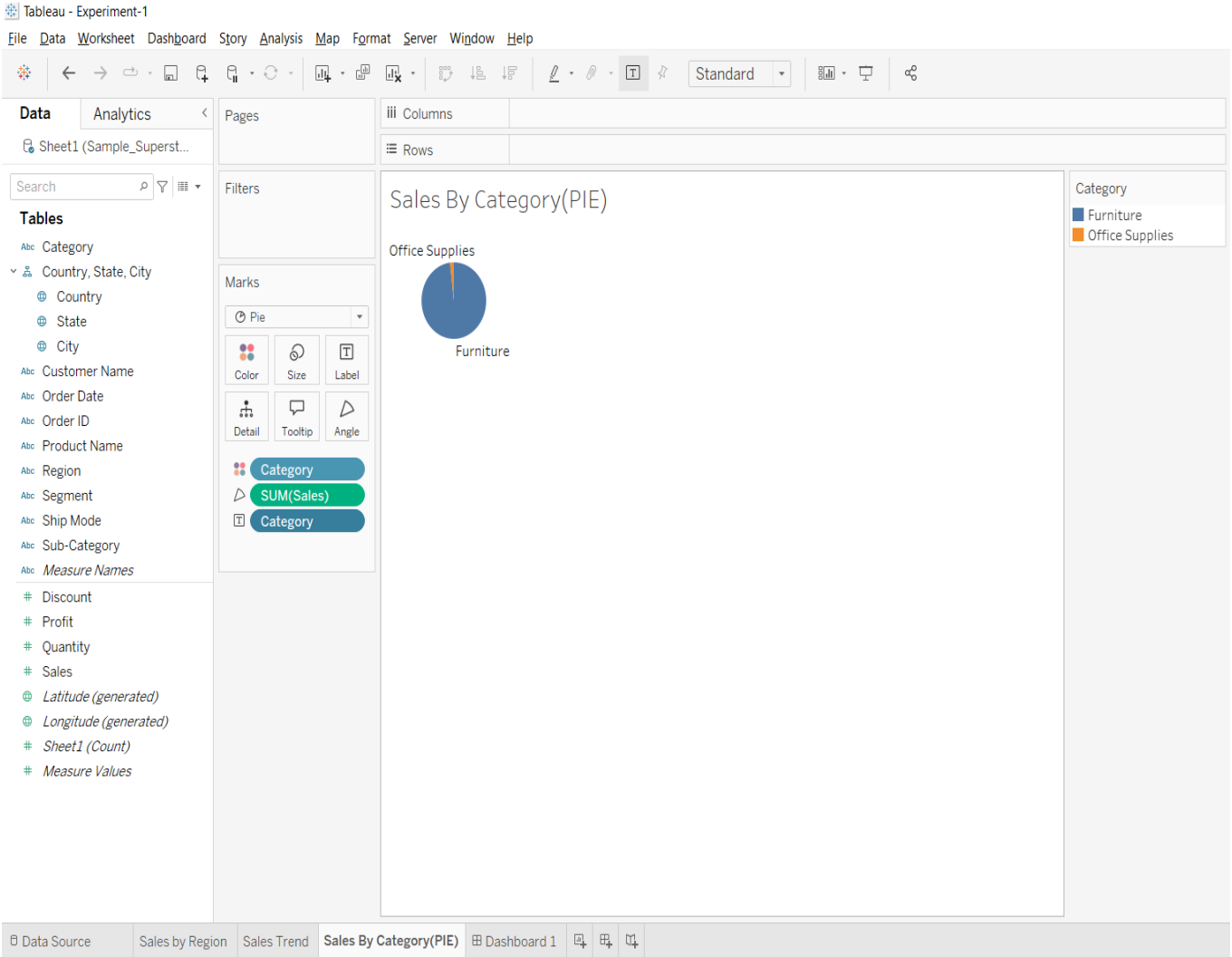
Sheet 2: Sales Trend Over Time

- Drag `Order Date` → Columns (right-click and choose Month or Continuous Month)
- Drag `Sales` → Rows
- Rename sheet to: `Sales Trend`



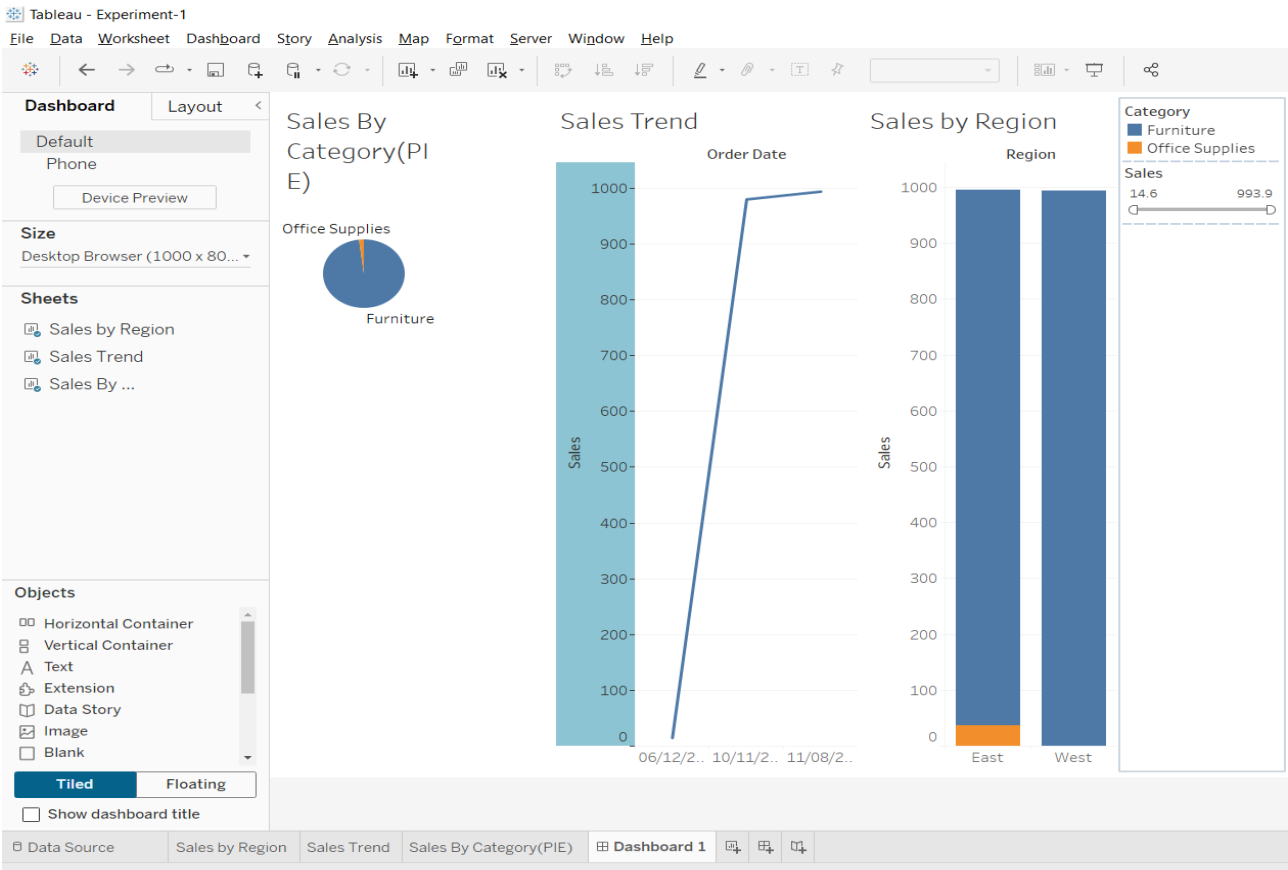
Sheet 3: Sales by Category (Pie Chart)

- Drag `Category` → Color
- Drag `Sales` → Angle
- On **Marks card**, change chart type to `Pie`
- Drag `Category` to Label
- Rename sheet to: `Sales Pie`



3. Create Dashboard

- Click on **New Dashboard** tab
- Drag `Sales by Region`, `Sales Trend`, and `Sales Pie` sheets into the canvas
- Adjust layout as needed
- Add **filters** (e.g., for Region or Year) if desired



4. Save as Packaged Workbook (`.twbx`)

- Click **File → Save As**
- Choose location and save as:
  `Experiment_1_Tableau_Superstore.twbx`

	Feature	`.twb` (Tableau Workbook)	`.twbx` (Tableau Packaged Workbook)
1.	**Format**	XML (text-based file)	ZIP archive containing `.twb` + data sources + resources
2.	**Contains Data?**	No — references data sources externally	Yes — contains the data (Excel, CSV, extracts, images)
3.	**File Size**	Smaller	Larger
4.	**Use Case**	When data is accessible from the same source (like a database or live file)	When sharing with others or publishing to Tableau Public
5.	**Dependencies**	Requires access to original data files	Self-contained (everything included)
6.	**Can be edited**	Yes — readable and editable in a text editor (XML)	Not easily editable (needs to be unpacked first)
7.	**Sharing**	Not ideal for sharing unless the recipient has the same data paths	Ideal for sharing or submitting as a project